

Double Roman Domination: A Survey

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Abstract: Since 2016, when the first paper of the double Roman domination appeared, the topic has received considerable attention in the literature. We survey known results on double Roman domination and some variations of the double Roman domination, and a list of open questions and conjectures is provided.

Keywords: domination; dominating functions; double Roman domination; variation of double Roman domination

MSC: 05C69, 05C35, 68R10, 68W01

1. Introduction

The concept of double Roman domination in graphs was introduced in 2016 by Beeler, Haynes and Hedetniemi in [1], based on Roman domination that was inspired by the strategies for defending the Roman Empire presented by Stewart [2] and ReVelle and Rosing [3]. Since then, almost 100 papers have been published on this topic including studies of double Roman domination and several variations. While it is an interesting fact that initially, modern studies of Roman domination [2–5] were motivated by a real problem from the fourth century, it is worth noting that further studies of the Roman domination of graphs and its variations are motivated by many other applications in the present time.

In the fourth century, Roman Emperor Constantine was confronted with a problem of how to defend his empire with a limited number of armies. The decision was taken to assign two types of military units to the empire provinces. While some units were able to move quickly from one location to another to respond to any attack, the second type was the provincial militia. These troops were permanently positioned in their home province. Emperor Constantine commanded that no legion should ever leave a province to protect another province, if doing so would leave the first province unprotected. Consequently, there were two armies in some provinces, and only local militia units were stationed in others. Thus, certain provinces had no permanent presence of an army and were protected by the armies from neighboring provinces. While the classical problem is still relevant in military operations research [6], we can also use it to model and solve the problems where time-critical assistance needs to be provided with some reserve. For example, first-aid medical assistance should never send its entire crew to answer a single emergency call. Studies of these types of domination problems are important for the optimization and efficient organization of emergency services.

A natural model is to define a graph where the provinces are vertices and two provinces are adjacent, so an army can move from one province to an attacked province in a reasonably short time. Recall that with Roman domination, one legion is required to defend any attacked vertex. What Beeler et al. propose in [1] is a stronger version of Roman domination that doubles the protection by ensuring that any attack can be defended by at least two legions. In Roman domination, at most two Roman legions are deployed at any one location. Clearly, the ability to deploy three legions at a given location provides a level



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of defense that is both stronger and more flexible, at less than the anticipated additional cost. It is called double Roman domination.

Following the reason mentioned above, understanding the double Roman domination problem and its variations may be crucial for positioning first aid stations, fire brigade stations, etc., at optimal positions. This may significantly improve public services at no extra cost. The double Roman domination is a natural generalization of Roman domination, where more than one army is supposed to be ready to respond. In the case of emergency services, at least two (or generally, k) teams may be needed to be quickly available in case of severe emergency calls. For more on other generalizations, see Section 4.14.

It is known that the decision version of the double Roman domination problem (MIN-DOUBLE-RDF) is NP-complete, even in cases when it is restricted to some special classes of graphs, for example to bipartite graphs, chordal (bipartite) graphs, planar graphs, circle graphs and to undirected path graphs [7–9]. A more detailed presentation follows in Section 6. Due to intractability of the problem, several avenues of research are of interest. For example, a number of studies regarding the complexity of the problem for special classes of graphs has been performed in recent years. The results include linear time algorithms for trees [10], for interval graphs and block graphs [9], for proper interval graphs [11] and for unicyclic graphs [7]. Another popular type of research are attempts to find exact values for the double Roman domination number of some graphs families, see Section 5.

Many papers have been published on double Roman domination since 2016, and the number is still growing rapidly. Therefore, it may be useful for the research community to provide a summary of the results and open problems. A detailed survey on what has been conducted on Roman domination and its variations can be found in five works of Chellali et al. [12–16]. Our intention was to collect all published works on the double Roman domination. We conducted a Google search and a scopus search with various keywords, and we believe that our list is close to complete. Our intention was also to keep the page-length of the survey within reasonable limits; therefore we only mention the results, either those that were emphasized by the authors or those that we find particularly interesting. While many problems and questions that arose in the initial studies have already been solved or answered, there are still some left unsolved, and in addition, the new papers open new perspectives that are accompanied by new open questions and research problems. Some of the open questions and problems are highlighted in the text, mostly those that have been emphasized by the authors of the original papers. The selection is thus necessarily not complete. Moreover, we do not claim that the problems emphasized are the only important open questions.

The rest of this paper is organized as follows. In Section 2, we summarize some basic definitions. In Section 3, the published results on double Roman domination are presented, while in Section 4 we provide definitions and results on different variations of double Roman domination. Known results on special classes of graphs are collected in Section 5. In Section 6 we survey published results on algorithmic complexity of the double Roman domination problem.

2. Preliminaries

A *directed graph* or *digraph* $G = (V, E)$ is a combinatorial object that is defined by two sets, the set $V = V(G)$ of *vertices*, and the set $E = E(G)$ of *edges*. An edge $e \in E$ connects its initial vertex $ini(e)$ and terminal vertex $ter(e)$. If no parallel edges exist, then we can write $e = (ini(e), ter(e))$ and $E = V \times V$. If, in addition, there is no loop in G , i.e., no edge with $ter(e) = ini(e)$, a graph is called a *simple graph*. Often, undirected graphs are considered. In this case, we define $e = \{ini(e), ter(e)\}$. In other words, an edge is a set of vertices. A loop is an edge that has one, and an edge that is not a loop has exactly two vertices. Most often, simple undirected graphs are studied, and in this case, the edges are given as pairs of vertices. Similarly, in simple digraphs, edges are ordered pairs of vertices.

We continue with some basic definitions of graph theory. For more detail and for ideas not defined here, we refer to textbooks [17–20]. Assume $G = (V, E)$ is a simple graph, i.e., a graph without loops and multiple edges. The *order* of G is $n = |V(G)|$, while the *size* of G is $m = |E(G)|$. The *open neighborhood* of a vertex v is the set $N(v) = \{u \in V(G) \mid uv \in E(G)\}$; its *closed neighborhood* is the set $N[v] = N(v) \cup \{v\}$. The *degree* of a vertex v equals $d(v) = |N(v)|$. The *maximum* and *minimum degree* of graphs G are denoted by $\Delta = \Delta(G)$ and $\delta = \delta(G)$, respectively. A vertex of degree one is called a *leaf*, while his neighbor is called a *support vertex*. A *r -regular graph* is a graph where each vertex has the same degree r . The *diameter* of a graph G , denoted by $\text{diam}(G)$, is the maximum distance between any two vertices in G . A graph G whose vertices and edges are subsets of another graph H is called a *subgraph* of G . If G has every possible edge inherited from H , it is an *induced subgraph*. The *complement* of a graph G will be denoted by $\bar{G}$. A graph containing no path of length three as an induced subgraph is called a *cograph*. Equivalently, in a cograph every nontrivial induced subgraph has at least one pair of twins—two vertices with the same neighborhoods. A *bipartite graph* is a graph with a set of vertices that could be decomposed into two disjoint sets such that no two vertices within the same set are adjacent. In a *chordal graph*, every induced cycle has exactly three vertices. A *block graph* is a type of undirected graph in which every block is a clique. (A *block* is an induced subgraph that is two-connected.)

A set $D \subseteq V(G)$ is called a *dominating set* of G if every vertex in $V(G) \setminus D$ has at least one neighbor in D . The cardinality of a minimum dominating set of G is called the *domination number* of G , $\gamma(G)$. A set S of vertices is called *independent* if no two vertices in S are adjacent. As usual, let P_n be the path of order n , C_n a cycle of length n and K_n the complete graph of order n . The star $K_{1,n}$ has one vertex of degree n and n leaves, $n \geq 1$.

Let $f : V(G) \rightarrow \{0, 1, 2\}$ be a function with the property that for every vertex $v \in V(G)$ with $f(v) = 0$, there exists a neighbor $u \in N(v)$ with $f(u) = 2$. Such a function will be called a *Roman dominating function*. The weight of a Roman dominating function equals the sum $\sum_{v \in V(G)} f(v) = f(V)$. The *Roman domination number* of G , denoted by $\gamma_R(G)$, equals the minimum weight of a Roman dominating function on G . A Roman dominating function on G with weight $\gamma_R(G)$ will be called a γ_R -function of G .

A *double Roman dominating function* (DRDF) on a graph $G = (V, E)$ is a function $f : V(G) \rightarrow \{0, 1, 2, 3\}$ with the properties that

- (1) Every vertex u with $f(u) = 0$ is adjacent to at least one vertex assigned three or at least two vertices assigned two;
- (2) Every vertex u with $f(u) = 1$ is adjacent to at least one vertex assigned two or three.

For an arbitrary subset $U \subseteq V(G)$, define the weight of f on U as $f(U) = \sum_{u \in U} f(u)$. Then, the weight of f equals $f(V) = \sum_{v \in V(G)} f(v)$. The *double Roman domination number* $\gamma_{dR}(G)$ of a graph G is the minimum weight of a double Roman dominating function of G . A DRD function f is called a γ_{dR} -function of G if $f(V) = \gamma_{dR}(G)$. A graph is called *double Roman* if $\gamma_{dR}(G) = 3\gamma(G)$. Let f be a double Roman dominating function on G . Define *double Roman dominating partition* (V_0, V_1, V_2, V_3) of the vertex set $V(G) = V_0 \cup V_1 \cup V_2 \cup V_3$, where $V_i = V_i^f = \{u \mid f(u) = i\}$. Clearly, any double Roman dominating partition exactly determines a double Roman dominating function and vice versa.

3. On Double Roman Domination

The study of the double Roman domination in graphs was introduced by Beeler et al. [1]. They proved that $2\gamma(G) \leq \gamma_{dR}(G) \leq 3\gamma(G)$ and noted that in a double Roman dominating function f of weight $\gamma_{dR}(G)$, no vertex needs to be assigned the value 1. Some other inequalities were proven: $\gamma_{dR}(G) \leq 2|V_1| + 3|V_2|$, and $\gamma_R(G) < \gamma_{dR}(G) \leq 2\gamma_R(G)$ with equality on the right side if and only if $G = \bar{K}_n$. Denote with $\mathcal{H}$ the family of connected graphs of order $n \geq 3$ that can be constructed from $n/4$ copies of P_4 by adding a connected subgraph on the set of centers of $\frac{n}{4}P_4$. For any connected graph G of order $n \geq 3$, $\gamma_{dR}(G) \leq \frac{5n}{4}$, with equality if and only if $G \in \mathcal{H}$.

Beeler et al. [1] asked for which classes of graphs or trees, there is $\gamma_{dR}(G) \leq n$. From the existing results in [21], we know that the following classes of graphs satisfy the inequality $\gamma_{dR}(G) \leq n$: (i) cycles C_n with $n = 2k$ or $n = 3k$ for some positive integer k ; (ii) paths P_n with $n = 3k$ for some positive integer k ; (iii) any graph with a universal vertex and $n \geq 3$; and (iv) any graph with a pair of twin vertices each of which dominates all other vertices and $n \geq 4$.

Problem 1. Characterize the graphs for which $\gamma_{dR}(G) \leq n$.

In the same paper [1], general upper bounds for $\gamma_{dR}(G)$ were studied. In particular, it was shown that with the exception of seven graphs, every connected graph G having minimum degree at least two satisfies $\gamma_{dR}(G) \leq \frac{6n}{5}$. Amjadi et al. [22] improved the upper bound by proving that $\gamma_{dR}(G) \leq \frac{8n}{7}$. In [23], Khoeilar et al. further improved this bound to $\gamma_{dR}(G) \leq \frac{11n}{10}$. They also provided an infinite family of graphs attaining their bound, hence it is the best possible in general case. In [24], it is proven that every connected graph G with minimum degree at least two and $G \neq C_5$ satisfies the inequality $\gamma_{dR}(G) \leq \lfloor \frac{13n}{11} \rfloor$. In [25], Kosari et al. show that this upper bound can be improved to $\frac{20n}{19}$ if G is restricted to connected graphs of order n with minimum degree at least two and no C_5 -cycle such that $G \notin C_7; C_{11}; C_{13}; C_{17}$. They also provide an infinite family of graphs attaining this bound.

In the case when a connected graph G has minimum degree at least three, the inequality $\gamma_{dR}(G) \leq \frac{9n}{8}$ was proven in [1]. In [26], Henning et al. improve the upper bound to $\gamma_{dR}(G) \leq \frac{97n}{88}$. Finally, it was shown that every connected graph G having minimum degree at least three satisfies the stronger inequality $\gamma_{dR}(G) \leq n$ [8]. It is not known whether this bound can be improved.

Problem 2 ([1,8]). Let G be a graph with minimum degree at least three. Is the upper bound $\gamma_{dR}(G) \leq n$ the best possible?

The double Roman domination number has been completely determined for paths and cycles by Ahangar et al. in [8],

$$\gamma_{dR}(P_n) = \begin{cases} n; & n \equiv 0 \pmod{3} \\ n+1; & n \equiv 1, 2 \pmod{3} \end{cases}, \quad \gamma_{dR}(C_n) = \begin{cases} n; & n \equiv 0, 2, 3, 4 \pmod{6} \\ n+1; & n \equiv 1, 5 \pmod{6} \end{cases}$$

They also show that the decision problem associated with double Roman domination is NP-complete for bipartite and chordal graphs, present some sharp bounds on the double Roman domination number and provide a characterization of graphs with small $\gamma_{dR}(G)$.

Anu et al. show in [21] that $\gamma_{dR}(G) + 2 \leq \gamma_{dR}(M(G)) \leq \gamma_{dR}(G) + 3$, where $M(G)$ is the Mycielskian graph of G . For any two positive integers a and b , they construct a graph G and an induced subgraph H of G such that $\gamma_{dR}(G) = a$ and $\gamma_{dR}(H) = b$ and found out that there is no relation between the double Roman domination number of a graph and its induced subgraph. They also study the impact of edge addition on the double Roman domination number and find an upper bound in terms of order and diameter.

Some new bounds of the double Roman domination numbers of graphs with given minimum degree and graphs of diameter two were given by Yue et al. in [27]. The authors show that the double Roman domination numbers of almost all graphs are at most n and obtain sharp upper and lower bounds for $\gamma_{dR}(G) + \gamma_{dR}(\overline{G})$. Furthermore, a linear time algorithm for the double Roman domination number of a cograph is given, and a characterization of the double Roman cographs is provided. Those results partially answer two open problems posed in [1].

Shao et al. in [28] show that a suitably altered discharging technique can be used on domination type problems. The general discharging approach for domination type prob-

lems is illustrated on a specific domination type problem, the double Roman domination on some generalized Petersen graphs. By applying this approach, they prove that

$$\gamma_{dR}(G) \geq \left\lceil \frac{3V(G)}{\Delta + 1} \right\rceil. \quad (1)$$

As examples, they also determine the exact values of the double Roman domination numbers of the generalized Petersen graphs $P(n, 1)$ and the double generalized Petersen graphs $DP(n, 1)$.

In [29], Volkmann initiated the study of the double Roman domatic number. A set $\{f_1, f_2, \dots, f_d\}$ of distinct double Roman dominating functions on G with the property that $\sum_{i=1}^d f_i(v) \leq 3$ for each $v \in V(G)$ is called a *double Roman dominating family (of functions)* on G . The maximum number of functions in a double Roman dominating family on G is the *double Roman domatic number* of G , denoted by $d_{dR}(G)$. Different sharp bounds on $d_{dR}(G)$ are given, and the double Roman domatic numbers of some classes of graphs are determined. Later [30], the same author continues the study of the double Roman domination and domatic numbers. In particular, he independently finds the lower bound (1) and determines the double Roman domination and domatic numbers of some more classes of graphs.

Hajibaba et al. in [31], present some relations of the double Roman domination number to an Italian domination number. Italian domination is a generalization of Roman domination, also called Roman 2-domination [32,33].

In [34], Jafari Rad et al. derive sharp upper and lower bounds on $\gamma_{dR}(G) + \gamma_{dR}(\overline{G})$ and $\gamma_{dR}(G)\gamma_{dR}(\overline{G})$. They also show that the decision problem for the double Roman domination number is NP-complete even when restricted to bipartite graphs and chordal graphs.

In [35], the concept of criticality with respect to various operations on graphs has been studied for double Roman domination in graphs. The authors characterize double Roman domination edge super critical graphs and give several characterizations for double Roman domination vertex (edge) critical graphs.

Yang et al. in [36], study properties of double Roman domination in graphs and find a class of double Roman graphs, the Cartesian product $C_{5m} \square C_{5n}$ for $m, n \geq 1$. They also give characterizations of trees with $\gamma_{dR}(T) = \gamma_R(T) + k$ for $k = 1, 2$.

Wu et al. in [37], introduced the term of double Roman domination stable graph. A graph G is *double Roman domination stable* if the double Roman domination number of G remains unchanged under removal of any vertex. The authors investigate properties of double Roman stable graphs, establish upper bounds on the order of double Roman stable graphs and characterize all double Roman stable unicyclic graphs.

In [38], Ouldrabah et al. note that $\gamma_{dR}(G) \leq 3\beta(G)$, for graph G without isolated vertices. Here, $\beta(G)$ denotes the *matching number* of G , the maximum size of a matching in G . Recall that a set $M \subseteq E$ is a *matching* if no two edges in M have a common vertex. Moreover, a descriptive characterization of block graphs G satisfying $\gamma_{dR}(G) = 3\beta(G)$ was given, and it was shown that the decision problem associated with $\gamma_{dR}(G) = 3\beta(G)$ is CO-NP-complete for bipartite graphs.

In [39], Volkmann provides Nordhaus–Gaddum type bounds on the double Roman domination number which improved the corresponding results given in [34]. For $G \notin \{K_n, \overline{K}_n, K_n - e, \overline{K}_n - e, C_4, 2K_2, P_4, C_5\}$, a graph of order n , $\gamma_{dR}(G) + \gamma_{dR}(\overline{G}) \leq 2n + 1$. In addition, a new lower bound on the double Roman domination number of trees (not a star) of order $n \geq 4$ with $\ell(T)$ leaves is established, $\gamma_{dR}(T) \geq \frac{n+8-\ell(T)}{2}$.

Very recently, Anu et al. [40] studied the impact of some graph operations, such as Cartesian product, addition of twins and corona with a graph, on the double Roman domination number.

4. Variations of Double Roman Dominating Functions

Although the first paper [1] on double Roman domination appeared quite recently, the basic definition of double Roman domination has already been altered in various ways,

in some cases motivated by particular potential applications. The extensions found in the literature include: total double Roman domination, signed double Roman domination, independent double Roman domination, restrained double Roman domination, maximal double Roman domination, inverse double Roman domination, perfect double Roman domination, global double Roman domination and double Roman domination subdivision as well as double Roman bondage number. A brief summary of results and open problems is provided in the next subsections.

4.1. Total Double Roman Domination

A *total double Roman dominating function* of a graph G without isolated vertices, written as TDRD-function (TDRDF), is a DRDF on G with the additional condition that the subgraph of G induced by the set $\{v \in V \mid f(v) \geq 1\}$ is isolated-free. The *total double Roman domination number* $\gamma_{tdR}(G)$ equals the smallest weight of a TDRDF on G .

This definition was first used in 2016 by Hao et al. [41], where some bounds on classes of graphs were given. For $n \geq 3$,

$$\gamma_{tdR}(C_n) = \left\lceil \frac{6n}{5} \right\rceil, \quad \gamma_{tdR}(P_n) = \begin{cases} 6; & n = 4 \\ \left\lceil \frac{6n}{5} \right\rceil; & \text{otherwise} \end{cases}$$

It was shown that for every isolated-free graph G , $\gamma_{dR}(G) \leq \gamma_{tdR}(G)$ and for any isolated-free graph G of order $n \geq 3$, $\gamma_{tdR}(G) = 4$ iff $\Delta(G) = n - 1$. In [42], it was shown that the decision problem for determining the total double Roman domination is NP-complete even for chordal and bipartite graphs. The authors provided sharp bounds on total double Roman domination number and gave some extension on the results given in [41]. Among other things, they showed that for any connected graph G of order $n \geq 2$, $\gamma_{tdR}(G) \leq \frac{3n}{2}$ with equality iff $G \in \{\text{corona of } H \mid H \text{ is a connected graph}\}$ and that $\gamma_{tdR}(G) \leq 2n - \deg(v)$ for any $v \in V(G)$.

The following conjecture regarding the upper bound for the total double Roman domination number $\gamma_{tdR}(G)$ appears in [42].

Problem 3 ([42]). *Is it true that for any graph G without isolated vertices of order n with $\delta(G) \geq 2$, $\gamma_{tdR}(G) \leq \frac{6n+4}{5}$?*

Later, several relations between the total double Roman domination number of a graph and other domination parameters were given by Hao et al. in [43]. In particular, for any graph G with no isolated vertex, $3\gamma_t(G) \geq \gamma_{tdR}(G) \geq \gamma_t(G) + \gamma(G)$. The lower bound is sharp for the disjoint union of copies of P_n . The upper bound is attained if and only if there exists a TDRDF f on G such that for each $v \in V(G)$ either $f(v) = 3$ or $f(v) = 0$. In addition, for any graph G with no isolated vertex, $\gamma_{tdR}(G) \leq 4\gamma(G)$ and $\gamma_{tdR}(G) \leq \frac{3}{2}\gamma_{dR}(G) - \frac{1}{2}$ with equality iff $\Delta(G) = n - 1$, $n \geq 3$. Moreover, $\gamma_{tR}(G) + 1 \leq \gamma_{tdR}(G) \leq 2\gamma_{dR}(G) - 1$. The lower bound is sharp for any graph with $\Delta = n - 1$, and the right-hand equality holds if $G = P_2$.

In [44], the study of the total double Roman domination number in digraphs was initiated, and its relationship to other domination parameters were shown. In particular, Amjadi et al. provided some bounds for the total double Roman domination number and determined the total double Roman domination number of some classes of digraphs.

Hao et al. show in [45] that for any connected graph G of order at least three, $\gamma_{tdR}(G) \geq \frac{3}{2}\rho^0(G) + 1$ with equality if and only if G is from the family of all graphs that can be obtained from a graph G of order n with $\Delta(G) = n - 1$ by adding a pendant edge to a vertex v of G where $d_G(v) = n - 1$. Here, $\rho^0(G)$ is the *open packing number* of G , the maximum cardinality of an open packing. A subset $S \subseteq V(G)$ is an *open packing* if the open neighborhoods of vertices in S are pairwise disjoint. Furthermore, for any tree T of order $n \geq 3$ with $\ell(T)$ leaves and $s(T)$ support vertices, $\frac{1}{5}(6n(T) + 3s(T)) \leq \gamma_{tdR}(T) \leq \frac{6}{5}(n(T) - \ell(T) + 2)$. The bounds are exact as examples of graphs are given achieving left or right equalities. For any

connected graph G of order n with minimum degree at least two, $\gamma_{tdR}(G) \leq \left\lfloor \frac{4n}{3} \right\rfloor$. The bound is sharp for C_n , $n \in \{3, 4, 5, 6, 7, 8, 11\}$.

The minimum total double Roman domination problem (MTDRDP) is to find a TDRDF of minimum weight of the input graph. In [46], Padamutham et al. showed that MTDRDP is polynomial time solvable for bounded treewidth graphs, chain graphs and threshold graphs. They designed a $3(\ln(\Delta - 0.5) + 1.5)$ -approximation algorithm for the MTDRDP, where Δ is the maximum degree of G . Even more, they showed that MTDRDP is approximation-hard for graphs with $\Delta = 5$.

4.2. Covering Total Double Roman Domination

A *covering total double Roman dominating function* (CTDRD function) on a graph G with no isolated vertex is a TDRD function for which $\{v \in V \mid f(v) \geq 1\}$ is a vertex cover set, or $V_0 = V_0^f = \{v \in V \mid f(v) = 0\}$ is an independent set. The covering total double Roman domination number $\gamma_{ctdR}(G)$ equals the minimum weight of a CTDRD function of G . The definition was initiated by Teymourzadeh et al. in [47], where the graphs G with small $\gamma_{ctdR}(G)$ were characterized. Furthermore, it was shown that the decision problem associated with CTDRD is NP-complete even when restricted to planar graphs with maximum degree at most four. It was shown that for every graph G without isolated vertices, $\gamma_{oitR}(G) < \gamma_{ctdR}(G) < 2\gamma_{oitR}(G)$, and for every tree T , $2\beta(T) + 1 \leq \gamma_{ctdR}(T) \leq 4\beta(T)$, where $\gamma_{oitR}(G)$ and $\beta(T)$ are the outer independent total Roman domination number of G , and the minimum vertex cover number of T , respectively. They also investigated the γ_{ctdR} of the corona of two graphs. Two research problems were proposed in [47]:

Problem 4 ([47]). Characterize the graphs G with large $\gamma_{ctdR}(G)$, covering the total double Roman domination number.

Problem 5 ([47]). For any graph G , obtain the lower and upper bound for $\gamma_{ctdR}(G)$.

4.3. Signed (Total) Double Roman Domination

The paper [48] initiated the research of signed double Roman domination in graphs. A *signed double Roman dominating function* (SDRDF) on a graph $G = (V, E)$ is a function $f : V(G) \rightarrow \{-1, 1, 2, 3\}$ such that

- (1) Every vertex v with $f(v) = -1$ is adjacent to at least two vertices assigned a 2 or to at least 1 vertex w with $f(w) = 3$;
- (2) Every vertex v with $f(v) = 1$ is adjacent to at least 1 vertex w with $f(w) \geq 2$;
- (3) $\sum_{u \in N[v]} f(u) \geq 1$ holds for any vertex v .

The weight of an SDRDF function is the value $\sum_{u \in V(G)} f(u)$; the minimum weight of an SDRDF is the *signed double Roman domination number* $\gamma_{sdR}(G)$. Ahangar et al. proved that the signed double Roman domination problem is NP-complete for bipartite and chordal graphs. For any tree T of order $n \geq 2$, it is shown that $\frac{-5n+24}{9} \leq \gamma_{sdR}(T) \leq n$. Trees attaining the lower and the upper bound are characterized.

The continuation in [49] gives several lower bounds on the signed double Roman domination number of a graph in terms of various graph invariants. In particular, the authors show that if G is a graph of order n and size m with no isolated vertex, then

$$\gamma_{sdR}(G) \geq \frac{19n - 24m}{9} \quad \text{and} \quad \gamma_{sdR}(G) \geq 4\sqrt{\frac{n}{3}} - n.$$

In [50], Amjadi et al. initiated the study of the signed double Roman k -domination number in graphs, motivated by previously defined signed Roman k -domination number in [51]. Let $k \geq 1$ be an integer. A *signed double Roman k -dominating function* (SDR k DF) on a graph G is a function $f : V(G) \rightarrow \{-1, 1, 2, 3\}$ such that

- (1) Every vertex v with $f(v) = -1$ is adjacent to at least 2 vertices with $f(w) = 2$ or to at least 1 vertex w with $f(w) = 3$;

- (2) Every vertex v with $f(v) = 1$ is adjacent to at least 1 vertex w with $f(w) \geq 2$;
 (3) $\sum_{u \in N[v]} f(u) \geq k$ holds for any vertex v .

The weight of an SDRkDF is $\sum_{u \in V(G)} f(u)$, and the minimum weight of an SDRkDF is the signed double Roman k -domination number $\gamma_{sdR}^k(G)$ of G .

Several questions were asked in [50] that may be of interest for further research of the signed double Roman k -domination.

Problem 6 ([50]). Find upper bounds on $\gamma_{sdR}^k(G)$ in terms of order of G and k .

Problem 7 ([50]). What can one say about the minimum and maximum values of $|V_{-1}|, |V_1|, |V_2|$ and $|V_3|$ for a γ_{sdR}^k -function $f = (V_{-1}, V_1, V_2, V_3)$ of a graph G ?

Problem 8 ([50]). Can one determine γ_{sdR}^k of grids, cylinders or tori?

Some basic properties and bounds for $\gamma_{sdR}^k(T)$ were presented by Amjadi et al. in [50]. For an r -regular graph G of order n such that $r \geq \left\lceil \frac{k-3}{3} \right\rceil$ and $n - r - 1 \geq \left\lceil \frac{k-3}{3} \right\rceil$ is

$$\gamma_{sdR}^k(G) + \gamma_{sdR}^k(\overline{G}) \geq \frac{4kn}{n+1}.$$

If n is even, then $\gamma_{sdR}^k(G) + \gamma_{sdR}^k(\overline{G}) \geq \frac{4k(n+1)}{n+2}$. Some other lower bounds for $\gamma_{sdR}^2(G)$ are given, and for a connected cubic graph G of order n and $k \leq 5$,

$$\frac{kn}{4} \leq \gamma_{sdR}^k(G) \leq \frac{13n}{8}.$$

In [52], the signed double Roman k -domination number of trees was investigated by Yang et al. The authors presented lower and upper bounds on $\gamma_{sdR}^k(T)$ for $2 \leq k \leq 6$ and classified all extremal trees.

The concept of signed total double Roman domination number was introduced in [53] by Shahbazi et al. A signed total double Roman dominating function (STDRDF) on a graph $G = (V, E)$ is a function $f : V(G) \rightarrow \{-1, 1, 2, 3\}$ such that: (i) every vertex v with $f(v) = -1$ is adjacent to at least 2 vertices assigned a 2 or to at least 1 vertex w with $f(w) = 3$, (ii) every vertex v with $f(v) = 1$ is adjacent to at least 1 vertex w with $f(w) \geq 2$, and (iii) $f(v) = \sum_{u \in N(v)} f(u) \geq 1$ holds for any vertex v . The weight of an STDRDF f is the value $w(f) = \sum_{u \in V(G)} f(u)$. The signed total double Roman domination number $\gamma_{sdR}^t(G)$ is the minimum weight of an STDRDF on G . Some bounds were also presented in [54], among others, for a connected graph G of order $n \geq 3$ and size m ,

$$\gamma_{sdR}^t(G) \geq \frac{11n - 12m}{3} \quad \text{and} \quad \gamma_{sdR}^t(G) \geq \left\lceil 3\sqrt{\frac{n}{2}} + \frac{1}{2} \right\rceil - n + 1$$

and for a tree of order $n \geq 3$, $\gamma_{sdR}^t(T) \leq \frac{4n}{3}$. Some recent results are presented in [55,56], where Amjadi et al. studied the SDRD/STDRD numbers of digraphs and presented lower and upper bounds for $\gamma_{sdR}(D)$ and $\gamma_{sdR}^t(D)$ in terms of the order, maximum degree and chromatic number of a digraph. In addition, the authors determined the SDRD/STDRD-numbers of some classes of digraphs.

Mahmoodi et al. in [57] initiated a term *twin signed double Roman domination* of a digraph. Let $D = (V, A)$ be a finite simple digraph. A function $f : V \rightarrow \{-1, 1, 2, 3\}$ is called a *twin signed double Roman dominating function* (TSDRDF) if: (i) every vertex v with $f(v) = -1$ has at least 2 in-neighbors assigned a 2 or at least an in-neighbor w with $f(w) = 3$, also at least 2 out-neighbors assigned a 2 or at least 1 out-neighbor w with $f(w) = 3$; (ii) every vertex v with $f(v) = 1$ is adjacent to at least an in-neighbor and an out-neighbor w with $f(w) \geq 2$; and (iii) $f(N^-[v]) \geq 1$ and $f(N^+[v]) \geq 1$ hold for any vertex v . The weight of a TSDRDF f is $\omega(f) = \sum_{v \in V(D)} f(v)$, the *twin signed double Roman*

domination number $\gamma_{sdR}^*(D)$ of D is the minimum weight of a TSDRDF on D . The authors present some bounds for $\gamma_{sdR}^*(D)$.

4.4. Independent Double Roman Domination

A DRDF $f = (V_0, V_1, V_2, V_3)$ is called an *independent double Roman dominating function* (IDRDF) if $V_1 \cup V_2 \cup V_3$ is an independent set. The *independent double Roman domination number* $i_{dR}(G)$ is the minimum weight of an IDRDF on G , and an IDRDF of G with weight $i_{dR}(G)$ is called an i_{dR} -function. By definition, we have $\gamma_{dR}(G) \leq i_{dR}(G)$. The concept of independent double Roman domination was introduced in [58] by Maimani et al. who show that the decision problem associated with $i_{dR}(G)$ is NP-complete for bipartite graphs. Later, in [59], some sharp bounds on the independent double Roman domination number were presented, and the relationship between independent double Roman domination and independent three-rainbow domination was investigated. Some other bounds are given also in [60], by Mojdeh et al.

In [61], Pour et al. showed that for every tree T of order $n \geq 3$,

$$i_{dR}(T) \leq n + \frac{s(T)}{2},$$

where $s(T)$ is the number of support vertices of T , improving the $5n/4$ -upper bound established in [59]. This bound is definitely better since every tree of order at least three cannot have more than half of its order as support vertices. Moreover, the trees T of order $n \geq 3$ with $i_{dR}(T) = \frac{5n}{4}$ were characterized. Every graph G satisfies $i_{dR}(G) \leq 3i(G)$, where $i(G)$ is the independent domination number and Rahmouni et al. in [62] give a characterization of all trees T with $i_{dR}(T) = 3i(T)$. Recently, in [63] Kheibari et al. show that for every tree T with diameter at least three, $i(T) + i_R(T) - \frac{s(T)}{2} + 1 \leq i_{dR}(T) \leq i(T) + i_R(T) + s(T) - 2$, where $i(T)$, $i_R(T)$ and $s(T)$ are the independent domination number, the independent Roman domination number and the number of support vertex of T , respectively.

4.5. Outer Independent Double Roman Domination

In [64], Ahangar et al. investigate the study of DRD functions $f = (V_0, V_1, V_2, V_3)$ for which V_0 is an independent set. This kind of function is called *outer independent double Roman dominating functions* (OIDRDFs), and the minimum weight of an OIDRDF of G is the *outer independent double Roman (OIDR) domination number* $\gamma_{oidR}(G)$. Various bounds on $\gamma_{oidR}(G)$ were provided, and it was shown that the corresponding decision problem is NP-complete on chordal and bipartite graphs. They also established Nordhaus–Gaddum bounds for $\gamma_{oidR}(G) + \gamma_{oidR}(\overline{G})$. Moreover, it has been shown that for any tree T of order $n \geq 3$, $\gamma_{oidR}(T) \leq 5n/4$ and the problem of characterizing those trees attaining equality was raised. In [65], by Rao et al., this problem was solved, and some additional bounds on the outer-independent double Roman domination number were given. In particular, for any connected graph G of order n with minimum degree at least 2 in which the set of vertices with degree at least 3 is independent, $\gamma_{oidR}(G) \leq 4n/3$. The following questions and problems were proposed in [64,65].

Problem 9 ([64]). Is it true that for any graph G on $p \geq 4$ vertices, $\gamma_{oidR}(G) \leq \frac{5p}{4}$?

Problem 10 ([64]). Find all graphs G , for which $\gamma_{oidR}(G) = 3\alpha_0(G)$, where $\alpha_0(G)$ is the minimum cardinality of a vertex cover of a graph G . (Vertex cover is a set of vertices that covers all the edges.)

Problem 11 ([65]). Find a sharp upper bound for the outer-independent double Roman domination number of connected graph G of order n with minimum degree.

At the same time, Mojdeh et al. [66] proved that the decision problem associated with $\gamma_{oidR}(G)$ is NP-complete even when restricted to planar graphs with maximum degree at most four. They also characterized the families of all connected graphs with small outer-independent double Roman domination numbers.

Later, in [67], characterizations of the families of all connected graphs with small outer independent double Roman domination numbers, and tight lower and upper bounds on this parameter were given. For a tree T , it was shown that this parameter is bounded from below by two times the vertex cover number of T plus one. Mojdeh et al. also proved that the decision problem associated with $\gamma_{oidR}(G)$ is NP-complete even when restricted to planar graphs with maximum degree at most four. Finally, they gave an exact formula for this parameter concerning the corona graphs.

In [68], Gao et al. investigate $\gamma_{oidR}(G)$ of regular graphs and give lower bounds on them. They present upper bounds on $\gamma_{oidR}(G)$ for torus graphs and determine the exact value of $\gamma_{oidR}(G)(C_3 \square C_n)$. It follows that $\gamma_{oidR}(G)(C_m \square C_n) \leq 5mn/4$ which verifies the open question of [64] (Problem 9) in this case.

In [69], the notion *outer independent signed double Roman domination* (OISDRD-number) was introduced by Ahangar et al., and it was shown that this problem is NP-complete for bipartite and chordal graphs. Exact values of the outer independent signed double Roman domination number γ_{sdR}^{oi} for paths and cycles were provided. Moreover, the authors showed that for trees T of order $n \geq 3$, $\gamma_{sdR}^{oi} \leq n - 1$, and they characterized extremal trees attaining this bound.

4.6. Restrained Double Roman Domination

In [70] by Mojdeh et al., the study of restrained double Roman domination was initiated. A *restrained double Roman dominating (RDRD) function* is a DRD function f such that the subgraph induced by V_0 , $G[V_0]$, has no isolated vertex. Recall that V_0 is the set of vertices with zero labels under f . The RDRD number $\gamma_{rdR}(G)$ is the minimum weight of an RDRD function f of G . An RDRD function f of a graph G with weight $\gamma_{rdR}(G)$ is called a $\gamma_{rdR}(G)$ -function. It was proved that the problem of computing this parameter is NP-hard. Mojdeh et al. gave an upper bound on the restrained double Roman domination number of a connected graph G in terms of the order of G and characterized the graphs attaining this bound. They also studied the restrained double Roman domination versus the restrained Roman domination and investigated the bounds for the restrained double Roman domination of trees and determined the trees attaining the exhibited bounds.

Gao et al. in [71] have shown that the decision problem associated with $\gamma_{rdR}(G)$ is NP-complete for chordal graphs. They also studied this graph parameter for the strong and direct products of two graphs.

In [72], Samadi et al. proved that the problem of computing $\gamma_{rdR}(G)$ is NP-hard even for planar graphs, but it is solvable in linear time when restricted to bounded clique-width graphs such as trees, cographs and distance-hereditary graphs. Relationships between $\gamma_{rdR}(G)$ and some well-known parameters such as restrained domination number, domination number or restrained Roman domination number are investigated. It was proven that $\gamma_{rdR}(T) \geq n + 2$ for any tree $T \neq K_{1,n-1}$ of order $n \geq 2$, and the family of all trees attaining the lower bound was characterized. There is also a characterization of graphs with small RDRD numbers.

Recently, Xi et al. in [73] characterize the graphs with small RDRD numbers and show the sharp bounds of $\gamma_{rdR}(G) + \gamma_{rdR}(\overline{G})$ for any connected graph G with order at least 3. A linear time algorithm for computing the RDRD-number of any cograph is presented. These results partially answer two of the open problems posed in [70] that we list below.

Problem 12 ([70]). Characterize the graphs G with small or large RDRD numbers.

Problem 13 ([70]). Provide nontrivial sharp bounds on $\gamma_{rdR}(G)$ for general graphs G or for some well-known families such as chordal, planar, triangle-free or claw-free graphs.

Problem 14 ([70]). The decision problem RDRD is NP-complete for general graphs. Provide families of graphs such that RDRD is NP-complete for them. Are there polynomial-time algorithms for computing the RDRD number of some well-known families of graphs, for instance, trees?

Problem 15 ([70]). What are the sufficient and necessity conditions for equalities in upper bounds $\gamma_{rdR}(T) \leq \left\lceil \frac{3n-1}{2} \right\rceil$, and $\gamma_{rdR}(T) \leq \frac{4n+2s-\ell}{3}$?

Problem 16 ([72]). Characterize trees T with $\gamma_{rdR}(T) = \gamma(T) + \gamma_r(T) + 1$. (γ_r denotes the restrained domination number.)

4.7. Double Roman Domination Subdivision Number

In [74], by Amjadi et al., the *double Roman domination subdivision number* $sd_{\gamma_{dR}}(G)$ of a graph G is defined, as the minimum number of edges that must be subdivided (each edge in G can be subdivided at most once) in order to increase the double Roman domination number. It was shown that the problem associated with $sd_{\gamma_{dR}}(G)$ is NP-hard, and some upper bounds on this parameter for arbitrary graphs were established.

4.8. Maximal Double Roman Domination

A double Roman dominating function $f = (V_0, V_1, V_2, V_3)$ on G is a *maximal double Roman dominating function* (MDRDF) if V_0 is not a dominating set of G . The *maximal double Roman domination number* $\gamma_{dR}^m(G)$ equals the minimum weight of an MDRDF of G . A $\gamma_{dR}^m(G)$ -function is an MDRDF of G with weight $\gamma_{dR}^m(G)$. This term was introduced in [75]. It is proved that the problem of determining $\gamma_{dR}^m(G)$ is NP-complete for bipartite, chordal and planar graphs. On the other hand, the problem is solvable in linear time for bounded clique-width graphs including trees, cographs and distance-hereditary graphs. Various relationships were also established relating to some domination parameters. For trees, it is shown that for every tree T of order $n \geq 4$, $\gamma_{dR}^m(T) \leq 5n/4$. All trees attaining the bound are characterized, and exact values of $\gamma_{dR}^m(G)$ for paths and cycles are given.

4.9. Inverse Double Roman Domination

In [76], a new type of inverse dominating function was introduced. If f is a DRDF on G with minimum weight $\gamma_{dR}(G)$, then its *inverse double Roman dominating function* (IDRDF) f' is a DRDF on G , such that $f'(v) = 0$, $\forall v \in S$, where $S = \{v \in V(G) \mid f(v) > 0\}$. The *inverse double Roman domination number* (IDRDN) of G , denoted by $\gamma_{idR}(G)$, is the minimum weight of such a function. D'Souza et al. obtained some bounds for the IDRDN of G and characterized the graphs having $\gamma_{idR}(G) = 3, 4, 5$ and higher. An approach for constructing graphs with the desired IDRDN was outlined.

4.10. Perfect Double Roman Domination

Introduced in [77] by Egunjobi et al., the *perfect double Roman dominating function* of a graph G is a function $f : V(G) \rightarrow \{0, 1, 2, 3\}$ satisfying the conditions that: (i) if $u \in V_0$, then u is either adjacent to exactly 2 vertices in V_2 and no vertex in V_3 or adjacent to exactly 1 vertex in V_3 and no vertex in V_2 ; and (ii) if $u \in V_1$, then u is adjacent to exactly 1 vertex in V_2 and no vertex in V_3 . The *perfect double Roman domination number* of G , denoted $\gamma_{dR}^p(G)$, is the minimum weight of a perfect double Roman dominating function of G . It was proven that if T is a tree of order $n \geq 3$, then $\gamma_{dR}^p(T) \leq 9n/7$. In addition, a family of trees of order n was given for which $\gamma_{dR}^p(T)$ approaches this upper bound as n goes to infinity.

4.11. Global Double Roman Domination

In [78], the global double Roman domination was proposed by Shao et al. A DRDF f is called a *global double Roman dominating function* (GDRDF) if f is also a DRDF of the complement $\bar{G}$ of G . The weight of a GDRDF is the sum of its function values over all vertices. The *global double Roman domination number* of G , denoted by $\gamma_{gdR}(G)$, is

the minimum weight of a GDRDF on G . Some properties of the global double Roman domination number were also obtained. Several research problems were proposed.

Problem 17 ([78]). Characterize the graphs G of order n with $\gamma_{gdr}(G) = 2n - 2$.

Problem 18 ([78]). Characterize the graphs G with $\gamma_{gdr}(G) = \gamma_{dr}(G) + t$ for each $t \in \{0, 1, 2, 3, 4, 5\}$. (Solved for trees in [79].)

Problem 19 ([78]). Characterize all connected graphs G achieving the bounds in inequality $\gamma_{gr}(G) \leq \gamma_{gdr}(G) \leq 2\gamma_{gr}(G)$.

Problem 20 ([78]). Characterize the trees T attaining the bounds in inequality $2\gamma_g(T) \leq \gamma_{gdr}(T) \leq 3\gamma_g(T)$. (Here, γ_g and γ_{gr} are global domination number and global Roman domination number, respectively.)

Hao et al. [79] established some inequalities between the global double Roman domination number $\gamma_{gdr}(G)$ and double Roman domination number $\gamma_{dr}(G)$ of graphs. They characterized the trees T with $\gamma_{gdr}(T) = \gamma_{dr}(T) + k$ for $k \in \{0, 1, 2, 3\}$, which partially answers Problem 18 above.

4.12. Double Roman Bondage Numbers of Graph

The *double Roman bondage number* of G , denoted by $b_{dr}(G)$, introduced by Jafari Rad et al. in [80], is the minimum cardinality among all edge subsets $B \subseteq E(G)$ such that $\gamma_{dr}(G - B) > \gamma_{dr}(G)$. The authors determined the double Roman bondage number in several families of graphs and presented some bounds for this parameter. They also studied the complexity issue of the double Roman bondage number and proved that the decision problem for the double Roman bondage number is NP-complete even when restricted to bipartite graphs.

4.13. Double Roman Reinforcement Number

By the definition of Amjadi et al. in [81], the *double Roman reinforcement number* $r_{dr}(G)$ of a graph G is the minimum number of edges that have to be added to G in order to decrease the double Roman domination number. The authors show that the decision problem associated to the double Roman reinforcement problem is NP-complete even when restricted to bipartite graphs, investigate the properties of double Roman reinforcement number in graphs and present some sharp bounds for $r_{dr}(G)$. They also give a characterization of trees with a double Roman reinforcement number greater than one.

4.14. $[k]$ -Roman Domination

$[k]$ -Roman domination has been defined by Ahangar et al. in [82]. For $k = 2$, the definition is equivalent to double Roman domination. Clearly, results for $[k]$ -Roman domination that hold for general k directly apply to double Roman domination. However, at present, we have only found one paper that provides results for general k . Haghparast et al. [83] consider the $[k]$ -Roman domination subdivision number, prove it is NP-hard and establish some properties and bounds.

Triple Roman domination (case $k = 3$) has been considered in [82,84], and quadruple Roman domination is studied in [85–87].

For completeness, we recall the definition of $[k]$ -Roman domination. Let $k > 1$ be an integer and f be a function that assigns labels from the set $\{0, 1, \dots, k+1\}$ to the vertices of G . The active neighborhood $AN(v)$ of a vertex $v \in V(G)$ with respect to f is the set of all vertices $w \in N(v)$ such that $f(w) > 1$. A $[k]$ -Roman dominating function on a graph G , abbreviated $[k]$ -RDF, is a function $f : V(G) \rightarrow \{0, 1, \dots, k+1\}$ such that for any vertex $v \in V(G)$ with $f(v) < k$, $f(N(v)) \geq |AN(v)| + k$. The $[k]$ -Roman domination number is the minimal weight of a $[k]$ -RDF.

We wish to note that $[k]$ -Roman domination should not be confused with some other generalizations of Roman domination. Alternative relevant answers to multiple attacks, or, multiple emergency calls, have been defined. Henning defines k -Roman domination in [4]. Later, the Roman k -domination [88] is studied by Kammerling et al. (Roman 3-domination is called double Italian domination [89], so perhaps Roman k -domination may be called $(k - 1)$ Italian domination.) Roman k -tuple domination is introduced in [90].

5. Double Roman Domination of Special Classes of Graphs

5.1. Trees

Trees are among the most popular graph classes. Usually, stronger results can be proved, in many cases nice characterizations exist and fast algorithms can be designed for solving various problems on trees. Therefore, it is natural to start research of a new concept by considering the trees. Beeler et al. [1] showed that $2\gamma(T) + 1 \leq \gamma_{dR}(T) \leq 3\gamma(T)$ for any non-trivial tree T . Two natural questions were asked in [1]:

- (1) Is it possible to construct a polynomial algorithm for computing the value of $\gamma_{dR}(T)$ for any tree T ? and
- (2) When is there equality in $2\gamma(T) + 1 \leq \gamma_{dR}(T)$?

In [10], Zhang et al. answer the first problem by giving a linear time algorithm to compute the value of $\gamma_{dR}(T)$ for any tree T . Moreover, characterizations of trees with $2\gamma(T) + 1 = \gamma_{dR}(T)$ and $\gamma_{dR}(T) + 1 = 2\gamma(T)$ were provided. The second question was completely answered in [8], namely $\gamma_{dR}(T) = 2\gamma(T) + 1$ if and only if $T \in \mathcal{T}$, where $\mathcal{T}$ the family of trees $T_{k,j}$, of order $k \geq 2$, where $k \geq 2j + 1, j \geq 0$, obtained from a star by subdividing j edges exactly once.

In [91], Mojdeh et al. characterize trees T with $\gamma_{dR}(T) = 3\gamma(T)$. The characterization of the double Roman trees with $\gamma_{dR}(T) = 3\gamma(T)$ is also given by Henning et al. in [26]. It is not clear who found the results first. Most likely, as many publications are in print at the same time, the work has been conducted independently, as perhaps holds in some other cases where the same results repeat as new.

In [92], Amjadi et al. show that if T is a tree, then $\gamma_{dR}(T) \leq 2\beta_2(T)$, and they characterize all trees attaining the equality. There, $\beta_2(G)$ is the 2-independence number, the maximum cardinality of a 2-independent set of G . Remember, a subset S of V is a 2-independent set of G if every vertex of S has at most 1 neighbor in S . Some improvement of upper and lower bounds for $\gamma_{dR}(T)$ considering the numbers of leaves and support vertices of a tree T were also given in [93] by Nazari-Moghaddam et al.

In [94], Ahangar et al. characterize all trees T with $\gamma_{dR}(T) = 2\gamma(T) + 2$. This, together with characterization of [10] provides partial answer to the next problem (Problem 21), for $k = 1$ and $k = 2$.

We conclude the section with two open questions from [91].

Problem 21 ([91]). Is there any family of graphs G , such that $\gamma_{dR}(G) = 2\gamma(G) + k$ where $2 \leq k \leq \gamma(G) + 1$? (For $k = 1$ and $k = 2$ see [10,94].)

Problem 22 ([91]). Is there any family of trees T , such that $\gamma_{dR}(T) = 2\gamma(T) + k$ where $2 \leq k \leq \gamma(T) + 1$?

5.2. Product Graphs

An upper and a lower bound for double Roman domination numbers on cardinal product of any two graphs was given in [95]. Klobučar et al. also determine the exact values of double Roman domination numbers on $P_2 \times G$ for many types of graph G . Furthermore, the double Roman domination number is found for $P_2 \times P_n, P_3 \times P_n$ and $P_4 \times P_n$. Upper and lower bounds are given for $P_5 \times P_n$ and $P_6 \times P_n$. This solves Problem 24 below in part.

For arbitrary graphs G and H , the cardinal product is the graph $G \times H$ which satisfies the following: Its vertex set is $V(G \times H) = V(G) \times V(H)$, and two vertices $(g, h), (g_0, h_0) \in V(G \times H)$ are adjacent if and only if g is adjacent to g_0 in G and h is adjacent to h_0 in H .

The cardinal product is one of the standard graph products, see [96]. It is natural to consider other standard products of graphs. For example, the grids and torus graphs are Cartesian products of paths and cycles.

In [36], it is observed that $\gamma_{dR}(C_{5m} \square C_{5n}) = 15mn$ implying that the Cartesian products $C_{5m} \square C_{5n}$ for $m, n \geq 1$ are double Roman graphs. Lower and upper bounds for a double Roman domination number of a Cartesian product of two graphs are given in preprint [40], where it is also observed that for the $2 \times n$ grid graph, $\gamma_{dR}(P_2 \square P_n) = \lfloor \frac{3n+4}{2} \rfloor$. The restrained double Roman domination number $\gamma_{rdR}(G)$ of strong and direct products of paths and cycles is considered by Gao et al. [71].

A more systematic study of double Roman domination of graph products may be an interesting avenue of future research. We formulate this main ideas as a couple of open problems.

Problem 23. Find tight bounds for the double Roman domination of graph product (strong, Cartesian or cardinal) in terms of double Roman domination and/or other graph invariant of the factors.

Problem 24. Find exact values of double Roman domination of graph product (strong, Cartesian or cardinal) for the products of (small) path or cycle with path P_n and/or cycle C_n , $n \in \mathbb{N}$. (γ_{dR} for product of paths P_2 , P_3 and P_4 with P_n is known [95].)

5.3. Generalized Petersen Graphs

The double Roman domination number for the generalized Petersen graphs $P(ck, k)$ for integer constants $c \geq 3$ was studied by Shao et al. [28]. They determined the exact value of $\gamma_{dR}(P(n, 1))$, while Jiang et al. [97] determined $\gamma_{dR}(P(n, 2))$. For more examples, see [28,97,98].

Clearly, as the generalized Petersen graph $P(n, k)$ is 3-regular and has $2n$ vertices, the bound (1) implies $\gamma_{dR}(P(n, k)) \geq \lceil \frac{3n}{2} \rceil$.

Gao et al. [98] determined the exact value of $\gamma_{dR}(P(n, k))$ for $n \equiv 0 \pmod{4}$ and $k \equiv 1 \pmod{2}$ and gave upper bounds for $\gamma_{dR}(P(n, k))$ in other cases.

Double Roman domination of families $P(ck, k)$ has also been studied for small k , including $c = 3, 4$, and 5 . For $k = 3$, exact values are known [99]:

$$\gamma_{dR}(P(3k, k)) = \begin{cases} 5k + 1, & k \in \{1, 2, 4\} \\ 5k, & \text{otherwise} \end{cases}$$

In contrast to $c = 3$, for $c = 4$ and $c = 5$, exact values are known only for some cases, while in other cases only bounds with small gaps are known [98,100].

$$6k \leq \gamma_{dR}(P(4k, k)) \leq \begin{cases} 6k, & k \equiv 1 \pmod{2} \\ 6k + 3, & \text{otherwise} \end{cases}$$

$$8k \leq \gamma_{dR}(P(5k, k)) \leq \begin{cases} 8k, & k \equiv 2, 3 \pmod{5} \\ 8k + 2, & \text{otherwise} \end{cases}$$

It follows that the only double Roman graph in the set of generalized Petersen graphs $P(5k, k)$ is $P(5, 1)$.

The general case $P(ck, k)$ was considered in [101] where it was proven that for $c \geq 3$, and $k \geq 3$,

if $c \equiv 0 \pmod{4}$ and k odd, then $\gamma_{dR}(P(ck, k)) = \frac{3}{2}ck$, and
 if $c \not\equiv 0 \pmod{4}$ and k odd, then

$$\frac{3}{2}ck \leq \gamma_{dR}(P(ck, k)) < \begin{cases} \frac{3}{2}(c + \frac{1}{2})k, & c \equiv 1, 3 \pmod{4}, \\ \frac{3}{2}(c + \frac{2}{3})k, & c \equiv 2 \pmod{4}, \end{cases}$$

and,
 for k even,

$$\frac{3}{2}ck \leq \gamma_{dR}(P(ck, k)) < \begin{cases} \frac{3}{2}c(k + \frac{1}{2}), & c \equiv 0 \pmod{4}, \\ \frac{3}{2}(c + \frac{1}{2})(k + \frac{1}{2}), & c \equiv 1, 3 \pmod{4}, \\ \frac{3}{2}(c + \frac{2}{3})(k + \frac{1}{2}), & c \equiv 2 \pmod{4}. \end{cases}$$

We conclude the subsection with some open problems.

Problem 25 ([97]). Characterize or at least find new examples of generalized Petersen graphs that are double Roman. (Several examples are already known, i.e., $P(n, 1)$ is a double Roman graph for any $n \not\equiv 2 \pmod{4}$ [28], $P(n, 3)$ for $n \equiv 0 \pmod{4}$ is double Roman [98].)

Problem 26 ([100]). Can the small gaps between lower and upper bounds for $\gamma_{dR}(P(5k, k))$ (and, also for $\gamma_{dR}P(4k, k)$) be resolved by finding and proving exact values?

Problem 27 ([100]). Find closed expressions or good lower and upper bounds for $\gamma_{dR}(P(6k, k))$.

5.4. Sierpiński Graphs

The study of the double Roman domination number of generalized Sierpiński graphs $S(G, t)$ was initiated by Anu et al. [102]. They obtained a bound for the double Roman domination number of $S(G, t)$ and also found the exact value of $\gamma_{dR}(S(K_n, 2))$.

Liu et al. in [103] propose a minimal dominating set for each $S(K_n, t)$ so that the exact values of its Roman domination numbers and double Roman domination numbers are known.

A double Roman domination number of generalized Sierpiński graph $S(G, 2)$, where G is a cycle C_n , $n \geq 4$, a complete bipartite graph $K_{1,q}$ or $K_{2,q}$, $q \geq 2$ and a bistar $B_{m,n}$, $m, n \geq 3$ was obtained by Varghese et al. [104].

5.5. Digraphs

Hao et al. in [105] initiated the study of the double Roman domination of digraphs. A double Roman dominating function on a digraph D with vertex set $V(D)$ is defined as a function $f : V(D) \rightarrow \{0, 1, 2, 3\}$ having the property that if $f(v) = 0$, then the vertex v must have at least 2 in-neighbors assigned 2 under f or 1 in-neighbor w with $f(w) = 3$, and if $f(v) = 1$, then the vertex v must have at least 1 in-neighbor u with $f(u) \geq 2$. The authors give several relations between the double Roman domination number of a digraph and other domination parameters such as Roman domination number, k -domination number and signed domination number. Moreover, various bounds on the double Roman domination number of a digraph are presented, and a Nordhaus–Gaddum type inequality for the parameter is also given.

The paper of Ouldrabah et al. [106] gives a descriptive characterization for some classes of digraphs satisfying $\gamma_{dR}(D) = 2(n - \Delta^+(D)) + 1$, where $\Delta^+(D)$ is the maximum out-degree of D . In addition, a descriptive characterization for digraphs D of order $n \geq 4$ for which $\gamma_{dR}(D) + \gamma_{dR}(\overline{D}) = 2n + 3$ holds.

Several relations between the double Italian domination number and other domination parameters such as double Roman domination number, Italian domination number and domination number, are also established in [89] by Volkmann.

Volkman in [107] gave the definition of double Roman domatic number of a digraph. A set $\{f_1, f_2, \dots, f_d\}$ of distinct double Roman dominating functions on D with the property that $\sum_{i=1}^d f_i(v) \leq 3$ for each $v \in V(D)$ is called a *double Roman dominating family (of functions)* on D . The maximum number of functions in a double Roman dominating family on D is the *double Roman domatic number* of D , denoted by $d_{dR}(D)$. He presents different sharp bounds on $d_{dR}(D)$ and determines the double Roman domatic number of some classes of digraphs.

6. On Algorithmic Complexity of Double Roman Domination

The double Roman domination number has been completely determined in paths and cycles [8] and has been shown to be linear-time solvable in trees [10] and P_4 -free graphs [27]. However, for a general graph, it is NP-complete to determine whether a graph has a DRDF of weight at most k [8].

Cai et al. [108] studied the integer linear programming (ILP) formulations for the DRDP and introduced several extra constraints to strengthen some ILP formulations. Additionally, an $H(2(\Delta + 1))$ -approximation algorithm for DRDP was proposed based on one ILP formulation.

In [9], Banerjee et al. strengthen the known NP-completeness of the decision version of Min-Double-RDF by showing that the decision version of Min-Double-RDF remains NP-complete for undirected path graphs, chordal bipartite graphs and circle graphs. They also present linear time algorithms for computing the double Roman domination number in proper interval graphs and block graphs and discuss the approximability of Min-Double-RDF, presenting a 2-approximation algorithm in 3-regular bipartite graphs.

Padamutham et al. in [109] show that the DRDP problem is NP-complete for star convex bipartite graphs and comb convex bipartite graphs and that $\gamma_{dR}(G)$ is obtained in linear time for bounded tree-width graphs, chain graphs and threshold graphs, a subclass of split graphs. The authors also propose a $3(1 + \ln(\Delta + 1))$ -approximation algorithm for the minimum double Roman domination problem.

In [11], Poureidi presents an algorithm to compute the double Roman domination number of a given proper interval graph $G = (V, E)$ in $\mathcal{O}(|V|)$ time.

Poureidi and Jafari Rad in [7] show that the decision problem associated with double Roman domination is NP-complete even when restricted to planar graphs and show that the problem of deciding whether a given graph is double Roman is NP-complete even when restricted to bipartite or chordal graphs. They also provide linear algorithms that compute the domination number and the double Roman domination number of a given unicyclic graph and a linear algorithm that decides whether a given unicyclic graph is double Roman.

In [46], Padamutham et al. show that the minimum total double Roman domination problem MTDRDP is polynomial time solvable for bounded tree-width graphs, chain graphs and threshold graphs. They design a $3(\ln(\Delta - 0.5) + 1.5)$ -approximation algorithm for the MTDRDP, and show that the same cannot have a $(1 - \delta) \ln |V|$ ratio approximation algorithm for any $\delta > 0$ unless $P=NP$.

Recently, Poureidi in [110] proposed an algorithm to compute the double Roman domination number of an interval graph G in $\mathcal{O}(m + n)$ time, answering a problem posed in [9]. He shows that the decision problem associated with the double Roman domination is NP-complete for split graphs and that the computational complexities of the Roman domination problem and the double Roman domination problem are different.

Bonomo-Braberman et al. in [111] prove that $[k]$ -Roman domination can be solved in $\mathcal{O}(kc|V(G)| + |E(G)|)$ time on graphs of bounded tree-width, for some constant c . Moreover, for constant values of k , the complexity is linear. It has been noted [111] that some of the variants defined for Roman and double Roman domination, such as perfect [77], independent [59], outer independent [64] and total [42], can be modeled by making slight modifications to the previous model.

One can ask many natural questions, here we recall only two.

Problem 28 ([108]). Given that optimization solver was not efficient in solving large sparse graphs, designing new exact methods or other approximation algorithms for DRDP is an interesting direction for future work.

Problem 29 ([109]). As the MINIMUM DOMINATION problem is APX-hard for bounded degree graphs, the intuition suggests that MR2DP and the MDRDP could be APX-hard. Hence, determining whether or not MR2DP and MDRDP are APX-hard for bounded degree graphs remains open.

7. Conclusions

In this paper we survey the known results that have been published on double Roman domination and its variations after it was first studied in 2016. As the graph domination and its many variations are still very popular [112], it is not a surprise that the Roman domination has been studied extensively in the last twenty years, see the recent surveys [12–16]. Similarly, in the last six years, the number of papers on double Roman domination have grown rapidly. Our survey thus includes one paper on double Roman domination published in 2017, and already nine in 2018. The number of publications (that we have found) is 12 in 2019, 23 in 2020, 10 in 2021 and 25 in 2022. Furthermore, there are some papers already dated 2023 and/or deposited in archives as preprints. Several open questions and interesting avenues for further research appear in the literature, so it is expected that the interest in this topic will grow in the future.

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